MCS-333	EMERGING DATABASE TECHNOLOGIES	L T P 3 1 0	4 Credits

Course Objectives:

CO1	Overview of emerging database applications and challenges
CO2	Data Management Issues in Social Networks and Network Computing Systems
CO3	Data Mining and Privacy Preserving Data Mining
CO4	RFID and Sensor Stream data management
CO5	Business Process and Workflow Management

MODULE I

Mobile Databases: Mobile computing architecture Mobile environment characteristics Data management issues.
Multimedia Databases: Nature of Multimedia data and applications Data management

MODULE II

Multimedia database applications

MODULE III

Object Database System: Abstract data types Objects identity and reference types Inheritance Database design for ORDBMS ODMG data model and ODL OQL.

References

1. *Raghu Ramakrishnan and Johannes Gehrke, "Database Management Systems", McGraw Hill Publications.*
2. *Ramez Elmasri & B.Navathe, "Fundamentals of* ***Database*** *Systems", V Ed., Addison Wesley, 2008.*
3. *H.F. Korth and A.Silberschatz, "Database system concepts", III Ed., McGraw Hill.*
4. *Jeffrey A. Hoffer, Mary Prescott, Heikki Topi, "Modern Database Management" (9th Edition), Prentice Hall.*

Course Outcome:

CO1	Explain the characteristics, architecture of database approach, describe the components, major functions of a database system and give examples of their use
CO2	Compare and contrast appropriate data models, including concepts in modeling notation and how they would be used
CO3	Explain the use of integrating OO properties with relational modeling
CO4	Explain the techniques used for data fragmentation, replication, evaluate simple strategies for executing a distributed query and explain how the two-phase commit protocol is used to deal with committing a transaction that accesses databases stored on multiple nodes
CO5	Familiarize with the related areas in databases and gaining familiarity with other popular databases used in the industry